ECE501 - DIGITAL IMAGE PROCESSING

SAR-Optical Feature Fusion for Crop-Type Classification

REPORT - WEEK 6

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GitHub Repository:
https://github.com/srushtibeng-svg/DIP_2026_Project19-SOFF_Group20.git

1) Work done this week

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- Reviewed the Germany and Uruguay datasets and their SAR inventory files.
- Identified 299 fields for Germany and 572 fields for Uruguay.
- Checked Sentinel-1 SAR image availability for the selected pilot fields using Google Earth Engine (GEE).
- Selected four pilot fields: two from Germany and two from Uruguay.
- Compared available Sentinel-1 acquisition dates and relative orbits for the pilot fields.
- Selected candidate relative orbits for further processing:

    - Germany: Descending relative orbit 168.
    - Uruguay: Descending relative orbit 68.
- Downloaded test SAR data using Google Earth Engine in batches to avoid download-size limitations.
- Generated and inspected VV and VH SAR data in QGIS. The downloaded test raster for the Germany rapeseed field contains 21 × 21 pixels, 20 bands, and a 10 m pixel size.
- Examined the yield-mask TIFF files for all four pilot fields and generated candidate polygons from their valid pixels.
- Opened the candidate polygons and corresponding yield-mask rasters in QGIS to check their spatial alignment.
- Investigated the raw dataset for actual field-boundary information. The JSON metadata contains centroid coordinates but does not provide polygon boundaries. No separate vector boundary files were found in the inspected raw ZIP dataset.

2) Work Planned for Next Week

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- Investigate and obtain the actual field-boundary data for Germany and Uruguay from the available dataset sources.
- Verify the candidate field footprints against actual field boundaries before using them for further analysis.
- Finalize the study regions for all four pilot fields.
- Continue Sentinel-1 VV and VH downloads for the selected fields using suitable batches.
- Extend the pilot processing towards full-field coverage once reliable field boundaries are available.
- Continue SAR data preparation, co-registration, and spatial verification in QGIS.

3) Results

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- Prepared and checked SAR inventory information for 871 fields across Germany and Uruguay.
- Identified four pilot fields for initial SAR processing.
- Selected candidate Sentinel-1 relative orbits based on image availability for the pilot fields.
- Successfully generated test SAR GeoTIFF files using Google Earth Engine.
- Verified the test raster properties in QGIS, including CRS, pixel size, dimensions, and band count.
- Generated candidate polygons from the valid pixels of the yield-mask rasters for all four pilot fields.
- Completed QGIS overlay checks for the four pilot fields.

Note: The current SAR test rasters cover small rectangular regions around the field centroids (210 m × 210 m), not the complete field boundaries. The polygons generated from yield masks are candidate yield-data footprints and have not yet been confirmed as actual field boundaries.

4) Challenges Faced

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The raw dataset does not contain separate vector boundary files such as SHP, GeoJSON, or GPKG in the inspected ZIP archive.

- The JSON metadata provides field centroid coordinates but does not contain the actual field polygon geometry.
- Yield-mask rasters show valid yield-data regions, but these cannot be treated as verified field boundaries.
- Downloading a large number of Sentinel-1 images in a single request exceeded Google Earth Engine's download-size limit. This was addressed by dividing the downloads into smaller batches.
- Full-field SAR processing is currently pending until reliable field-boundary information is obtained.

5) References:

1. Google Earth Engine – Sentinel-1 GRD Dataset: https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD
2. Raw Germany and Uruguay dataset files and metadata provided for the project.
3. QGIS – Geographic Information System: https://qgis.org/